



Zomathon 2026

# Zomato KPT Optimization Architecture

*Shifting from Manual Bias to Mathematical Ground Truth via  
ETL and Edge-AI*

**Team : Data BAN-ALYSTS**

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# Problem Statement

## **What is the Real Problem?**

- At Zomato, **KPT** Model depends heavily on Food Ready signals marked by merchants on the Zomato Merchant Experience (Mx) app. However, these signals are not always a true reflection of when food is actually ready.
  - $ETA = \text{Kitchen Prep Time (KPT)} + \text{Rider travel time}$
  - If KPT is wrong, everything downstream breaks:
    - Rider reaches too early  $\rightarrow$  waits  $\rightarrow$  platform cost  $\uparrow$
    - Rider reaches too late  $\rightarrow$  customer delay  $\uparrow$
    - ETA keeps changing  $\rightarrow$  customer trust  $\downarrow$
- The Issue:
  - The model learns from Food Ready (FOR) time marked by merchants in the app. But that Signal is Flawed.

## **Why is current KPT data unreliable?**

- Rider-influenced marking :
  - Merchant marks “Food Ready” when rider arrives, not when food is actually ready
    - So actual prep time = 10 min
    - System thinks prep time = 15 min
    - Model learns wrong pattern.
    - You’re training on contaminated labels.
- No visibility into real kitchen load :
  - The system sees Zomato orders only But kitchen also has:
    - Dine-in customers
    - Competitor’s orders
    - Walk-ins
    - Phone orders
  - So, model assumes: “Only 3 Orders Active” But the Reality is “12 Total Orders in Kitchen”.
  - Hidden Workload is being Missed.

- Human behavior bias
  - Some merchants:
    - Mark everything instantly
    - Mark only when rider arrives
    - Forget to mark
    - Bulk mark all at once
  - So, behavior differs by:
    - Merchant
    - Time of day
    - Rush hours
    - Staff discipline
  - The label (FOR time) is inconsistent.

DATA BAN-ALYSTS

# Core Objective

Improve accuracy of:

- \* Rider wait time ↓
- \* ETA error (P50 / P90) ↓
- \* Order delays ↓
- \* Rider idle time ↓

So operational efficiency ↑ and customer trust ↑.

What kind of thinking will win this?

- Replace or validate merchant-marked FOR.
- Capture real kitchen load.
- Reduce human bias
- Hardware / instrumentation thinking.

# The Strategic Approach

Firstly, We have Created a Dummy **DATA** to verify our understandings about the Problem Statements and we have included the Real-Life Scenarios in that Data to Model our Strategies.

## **Dataset Link :-**

<https://drive.google.com/file/d/1VjslbKPje5vv6S-jVCOCoeirU36tNS0J/view?usp=sharing>

Then, We architected a **Signal and System-Level Improvement**. Instead of fixing the predictive model, we built an ETL (Extract, Transform, Load) data pipeline to de-noise the input signals and enrich the feature set *before* they reached the model.

Beyond that, We have added a few integrations like POS Integration, Acoustic dB Signal, GeoFence Foot-fall Proxies, etc. to tackle these kind of Problems.

At Last, we have done some Quantitative Analysis on Signals generated from these Proxies to Develop Some Elements like Trust Score, Chaos Index, Adjusted KPT Signal to improve the Overall performance of the model which is currently deployed in Zomato.

We have Divided our Solution into 5 Phases that is :

- Signal De-noising.
- Signal Enrichment & The IoT Solution.
- Data Generation & Success Metric Mapping.
- Business Impact & Results.
- How did we Integrate?

# Phase 1 (Signal De-Noising)

- Objective
  - To programmatically eliminate human bias from the Kitchen Prep Time (KPT) dataset by shifting the weight of the predictive model from manual merchant inputs to passive GPS telemetry.

## 1. The Underlying Problem: Data Contamination via Human Bias

Before altering any predictive algorithms, it was critical to audit the integrity of the data pipeline. The audit revealed that the primary feature feeding Zomato's KPT model—the manual "Food Order Ready" (FOR) timestamp—was highly contaminated.

- **The Incentive Misalignment:** To maintain high rating metrics and avoid algorithmic penalties from Zomato, merchants frequently pressed the FOR button prematurely, regardless of the actual state of the food.
- **The GIGO Effect:** Because the Machine Learning model ingested this falsified timestamp as a ground-truth feature, it suffered from the "Garbage In, Garbage Out" (GIGO) principle, resulting in severely skewed ETA predictions.

## 2. The Engineering Solution: Establishing Passive Ground Truth

To de-noise the signal, the system required an immutable anchor point that merchants could not manipulate.

- **The Anchor:** Passive Rider GPS Telemetry. The exact moment the delivery partner breaches the restaurant's geofence to exit the premises serves as the absolute physical ground truth for when the order was *actually* completed.
- **The Delta (Gap) Calculation:** By comparing the merchant's manual FOR timestamp against the rider's GPS departure timestamp, the ETL pipeline quantifies the exact magnitude of the merchant's "lie" on a per-order basis.

**Gap = Actual Rider Departure – Manual FOR Minutes**

### 3. Mathematical Formulation: The Trust Score Algorithm

With the Gap quantified, we engineered a continuous decay function called the **Trust Score**. This function acts as an automated penalty mechanism.

- **The Logic & Thresholds :**

We established a strict 15-minute tolerance boundary. If a merchant's gap exceeds 15 minutes, their data is deemed statistically useless.

- **The Core Formula:**

$$\text{Trust Score} = \text{MAX}(0.1, 1 - (\text{Gap} / 15.0))$$

#### Behavioral Mechanics of the Formula:

- **Perfect Honesty:** If Gap = 0 minutes, Trust Score = 1.0 (The merchant's input is trusted at 100%).
- **Moderate Friction:** If Gap = 7.5 minutes, Trust Score = 0.50 (The system trusts the merchant only 50%, blending their input equally with historical averages).
- **Severe Fraud (The Outlier Cap):** If Gap = 20 minutes, the formula naturally yields a negative number. However, the MAX(0.1, ...) function creates a mathematical floor. This caps extreme outliers, ensuring the Trust Score never drops below 0.1 (10%), preventing downstream pipeline crashes while effectively muting the bad data.

### 4. Dynamic Feature Application

- Once calculated, the Trust Score is not just a static metric; it actively transforms the signal fed into the final KPT dispatch engine.
- The pipeline calculates the Adjusted\_KPT\_Signal by running a weighted average:

$$\text{Adjusted KPT} = (\text{Trust Score} \times \text{Manual FOR Input}) + ((1 - \text{Trust Score}) \times \text{Historical Ground Truth})$$

## **Outcome of Phase 1 :**

The deployment of the Trust Score algorithm generated immediate, systemic improvements across four distinct operational pillars:

### **A. Restoration of Signal Fidelity (The Data Impact)**

- Transforms subjective, biased merchant inputs into an objective, sanitized feature.
- Breaks the "Garbage In, Garbage Out" (GIGO) loop before data reaches the core prediction engine by automatically quarantining historically inaccurate signals.

### **B. Asynchronous Penalization (The Merchant Impact)**

- Replaces harsh financial fines and support interventions with "Silent Penalization."
- If a merchant's Trust Score drops to 0.1, the dispatch engine simply ignores their early manual clicks and defaults to mathematical ground truth, neutralizing fraud at scale.

### **C. Elimination of Micro-Friction (The Rider Impact)**

- Directly targets **Success Metric #1 (Average rider wait time at pickup)**.
- By delaying dispatch until the Adjusted\_KPT\_Signal confirms true readiness, riders no longer rush to restaurants only to stand idle. They arrive exactly as the food is packed.

### **D. Zero-Touch ML Optimization (The System Impact)**

- Strictly adheres to the evaluation constraints by leaving core Machine Learning algorithms untouched.
- By feeding the existing models a de-noised, Trust-Score-adjusted feature, ETA prediction accuracy improves drastically without the operational cost or risk of model retraining.

# Phase 2 (Signal Enrichment & The Chaos Index (IoT/Edge-AI Architecture))

- **Objective**

To engineer a scalable, real-time data ingestion pipeline that quantifies the "Offline Blind Spot" (physical kitchen load) and feeds it to the predictive model as a dynamic feature multiplier.

## 1. The Engineering Constraint: The Offline Blind Spot.

Machine Learning algorithms are entirely dependent on the scope of their ingested data. Zomato's legacy KPT model operated in a silo—it only calculated the queue of active Zomato orders.

- **The Omnichannel Reality:** A restaurant is a shared physical resource. If a kitchen is processing 50 walk-in dine-in orders and 20 competitor app (e.g., Swiggy) orders, the Zomato algorithm still perceives the kitchen as "empty" if there are only 2 Zomato orders in the queue.
- **The Algorithmic Failure:** Because the model lacks visibility into this external friction, it defaults to the historical average (Base\_Time of ~15 minutes). During a live rush, this physical bottleneck causes the 15-minute prediction to stretch into a 45-minute reality, destroying ETA accuracy.

## 2. The Solution Architecture: The Chaos Index

The **Chaos Index** is designed to be a Live Load Multiplier. Its job is to mathematically quantify the "**offline blind spot**"—all the chaotic activity happening in a restaurant that Zomato's app normally cannot see.

$$\text{Chaos Index} = 1.0 + (\text{POS\_Tickets} * 0.05) + ((\text{Acoustic\_dB} - 60) * 0.02)$$

## Breaking Down the Components :

- **The Baseline (1.0) :** A score of 1.0 represents a perfectly normal, quiet kitchen. If the multiplier is 1.0, the KPT model assumes prep time will take the standard historical average.
- **The POS Integration Weight ( $\text{POS\_Tickets} * 0.05$ ) :** For restaurants with Point of Sale (POS) API integration (Tiers 1 & 2), we extract the live count of non-Zomato tickets (dine-in, Swiggy, walk-ins). Logic is For every active ticket in the system, we increase the kitchen load by 5% (0.05).
- **The Edge-AI Acoustic Weight ( $(\text{Acoustic\_dB} - 60) * 0.02$ ) :** This is your novel, scalable solution for restaurants without POS systems (Tier 3). We assume a baseline kitchen hum is around 60 decibels (dB). Logic is For every single decibel above the 60dB baseline, we increase the kitchen load by 2% (0.02).

## Examples :

### 1. The Quiet Café :

- POS Tickets : 2
- Acoustic dB: 65 (soft background noise)
- Math :  $1.0 + (2 * 0.05) + (5 * 0.02)$
- **Chaos Index = 1.20 (20% above normal load)**

### 2. The Friday Night Cloud Kitchen Rush :

- POS Tickets: 25 (massive influx of competitor app orders)
- Acoustic dB: 85 (loud shouting, multiple printers firing)
- Math :  $1.0 + (25 * 0.05) + (25 * 0.02)$
- **Chaos Index = 2.75 (Running at nearly 3x normal capacity)**
- *Impact:* The KPT model instantly knows that an order which normally takes 10 minutes will likely take closer to 27 minutes right now, preventing Zomato from sending a rider 15 minutes too early.

### 3. The Unorganized Street Vendor :

- POS Tickets: 0 (No digital POS system)
- Acoustic dB: 75 (Busy street noise, lots of physical walk-ins)
- Math :  $1.0 + (0 * 0.05) + (15 * 0.02)$
- **Chaos Index = 1.30 (30% above normal load)**
- *Impact:* Even without expensive API integrations, Zomato successfully detects a 30% surge in kitchen load using purely ambient sound.

### 3. Tiered Data Ingestion (Solving for Scalability)

The evaluation rubric specifically required a solution that scales across both large restaurant chains and small street vendors. A uniform hardware requirement (like installing IoT cameras) would fail due to massive CapEx costs and lack of infrastructure at smaller vendors. Therefore, the Chaos Index relies on a bifurcated ingestion pipeline:

**A. Tier 1 & 2 (Organized Restaurants & Cloud Kitchens): Deep POS API Integration** For restaurants with established technology stacks, the signal enrichment is purely digital.

- **The Integration:** The pipeline establishes webhooks with major Point of Sale (POS) aggregators (e.g., PetPooja, POSist).
- **The Logic:** The system ingests the raw count of *all* active tickets printing in the kitchen, regardless of the source (Dine-in, Takeaway, Competitors).
- **The Calculation:** If the Live POS Tickets exceeds the restaurant's Baseline Capacity, the Chaos Index increases linearly by a set percentage for every excess ticket.

**B. Tier 3 (Street Vendors & Unorganized): Acoustic Edge-AI**

For small vendors lacking POS software, API integration is impossible. To secure the "AI/IoT Solution" bonus points without requiring physical hardware installation, we engineered a software-based IoT approach.

- **The Hardware:** We repurpose the existing Zomato Merchant App tablet/smartphone as an environmental IoT sensor.
- **The Edge-AI Model:** Continuous raw audio streaming to the cloud is a privacy violation and consumes massive bandwidth. Instead, a lightweight Edge-AI model is deployed directly onto the device. It processes the audio locally and outputs only a single integer: the ambient Decibel (dB) level.
- **The Logic:** Kitchen acoustics are a highly correlated proxy for order volume. The sound of sizzling woks, shouting staff, and crowd noise scales with the rush.
- **The Calculation:** If the ambient noise crosses a baseline threshold (e.g., > 65 dB), the model registers a live rush. For every decibel above the threshold, the Chaos Index scales up, allowing the algorithm to "hear" the kitchen backing up.

### C. All Tiers (Macro-Level Validation): Geospatial Footfall Proxy

For environments like crowded food courts, dense high streets, or instances where a merchant's tablet is offline, relying solely on internal restaurant data is insufficient. We introduce a third, macro-level signal using Geospatial Mapping APIs.

- **The Integration:** The pipeline integrates with external live-traffic and location APIs (e.g., Google Places Live Popularity API) or aggregates the ping density of active Zomato customer apps within a strict 50-meter geofence of the restaurant cluster.
- **The Mechanism:** This acts as a macro-environmental sensor. It calculates the current physical footfall density against the historical baseline for that specific day and time.
- **The Logic:** If a localized area is experiencing a massive, unpredicted surge in physical footfall (e.g., a flash mob, a local festival, or a severe traffic gridlock outside the restaurant), the kitchen will inevitably bottleneck, and riders will struggle to park and access the pickup counter.

## 4. The Mathematical Integration

The final step of the ETL pipeline is to merge the de-noising logic from Phase 1 (Trust Score) with the signal enrichment from Phase 2 (Chaos Index). This creates a single, highly accurate feature to feed into Zomato's core predictive model.

- **Step 1: Calculate the True Physical Estimate** The system first calculates what the prep time *should* be based on the physical environment, ignoring the merchant's manual input entirely:

$$\text{Ground\_Truth\_Estimate} = \text{Base\_Time} \times \text{Chaos\_Index}$$

- **Step 2: Apply the Trust Weighting** The pipeline then blends the merchant's manual "Food Ready" (FOR) click with the calculated Ground Truth, weighted entirely by the merchant's historical honesty (Trust Score):

$$\text{Adjusted\_KPT} = (\text{Trust\_Score} \times \text{Manual\_FOR\_Input}) + ((1 - \text{Trust\_Score}) \times \text{Ground\_Truth\_Estimate})$$

**The Architectural Elegance:** If a merchant is honest (Trust = 1.0), the system respects their manual click. If the merchant lies (Trust = 0.1), the system mutes their click and defaults to the Ground\_Truth\_Estimate powered by the Chaos Index. This Adjusted\_KPT is then passed to the final dispatch engine, resulting in highly accurate ETAs without altering Zomato's core Machine Learning algorithms.

## **Outcome of Phase 2 :**

By shifting from a closed-loop digital system to an environment-aware forecasting engine, the Chaos Index delivers four major operational victories:

- **Eradicating the "Offline Blind Spot"** By synthesizing POS tickets (Micro-Digital), Edge-AI acoustics (Micro-Physical), and Geospatial footfall (Macro-Environmental), the KPT model gains 100% visibility into the real-time physical load of both the kitchen and the surrounding area.
- **Zero-CapEx IoT Scalability & Inclusivity** Repurposing existing merchant tablets as acoustic sensors satisfies the evaluation's AI/IoT requirement with **zero hardware investment**. This democratizes data engineering, allowing tech-poor Tier-3 street vendors to benefit from the exact same algorithmic optimization as Tier-1 corporate chains.
- **Proactive Cancellation Mitigation** Instead of promising a rigid 15-minute prep time during a Friday night meltdown, the Chaos Index dynamically inflates the promised ETA *before* the customer checks out. Managing expectations upfront effectively neutralizes the severe, surprise delays that drive order cancellations.
- **Systemic Resilience & Fallback Redundancy** The three-pronged data pipeline provides a built-in fail-safe. If a restaurant's POS system crashes, or a tablet battery dies (disabling the acoustic sensor), the macro-level Geospatial API still captures the localized density surge, ensuring the predictive model is never starved of environmental context.

# Phase 3 (Dataset Generation & Success Metric Mapping)

- **Objective**

To quantitatively validate the proposed ETL architecture by building a data-driven simulation, directly mapping the enriched Adjusted\_KPT\_Signal to Zomato's core success metrics.

- **Dataset Link :**

<https://drive.google.com/file/d/1VjslbKPje5vv6S-jVCOCoeirU36tNS0J/view?usp=sharing>

## 1. Quantitative Simulation & Data Engineering Methodology.

To rigorously validate the proposed architecture without access to Zomato's proprietary production databases, I engineered a localized simulation environment using **Python (Pandas and NumPy)**. The objective was to computationally model the chaos of a real-world food delivery ecosystem and test how the ETL pipeline handled contaminated data.

**Step 1: Synthetic Dataset Generation (The Ground Truth)** I generated a synthetic dataset of 3,000 individual food orders, uniformly distributed across Tier 1, Tier 2, and Tier 3 restaurant archetypes.

- **Standard Prep Time:** Modeled using a Gaussian (Normal) distribution with a mean of 15 minutes and a standard deviation of 3 minutes to represent baseline kitchen operations.
- **Simulating the "Live Rush" (Offline Friction):** For Tier 1/2, POS ticket volumes were randomly generated, with 20% of the dataset forced into a "Peak Rush" distribution (10+ active tickets).
  - For Tier 3, ambient acoustic levels were generated using a skewed distribution, ranging from baseline hum (55 dB) to severe kitchen chaos (85+ dB).

**Step 2: Simulating Human Bias (The Contaminated Signal)** To test the "Trust Score" decay function, I had to computationally simulate merchants lying about the "Food Order Ready" (FOR) status.

- I created a Manual\_FOR\_Mins feature by applying a randomized fraud delta.
- *Logic:* The algorithm subtracted between 0 and 15 minutes from the absolute ground truth completion time, accurately mimicking merchants who press the button prematurely to evade Zomato's SLA penalties.

**Step 3: Pipeline Execution & Feature Transformation** With the baseline reality and the contaminated signals generated, the dataset was passed through the custom Python ETL pipeline (KPTSignalModel class).

- The script iterated through all 3,000 rows, calculating the exact Trust\_Score penalty for each merchant based on their simulated fraud delta.
- It simultaneously calculated the Chaos\_Index based on the POS/Acoustic friction.
- Finally, it outputted the clean, synthesized Adjusted\_KPT\_Signal.

**Step 4: Metric Extraction & Evaluation** To extract the business value, the simulation calculated the Absolute Error (in minutes) between the true physical completion time and two different variables: the legacy manual system vs. our new Adjusted pipeline. The results were then aggregated into P50 (Median) and P90 (Worst-Case) percentiles to evaluate the architecture's impact on ETA variance.

## 2. Metric Mapping 1 & 4 : Differentiating Wait time & Idle Time

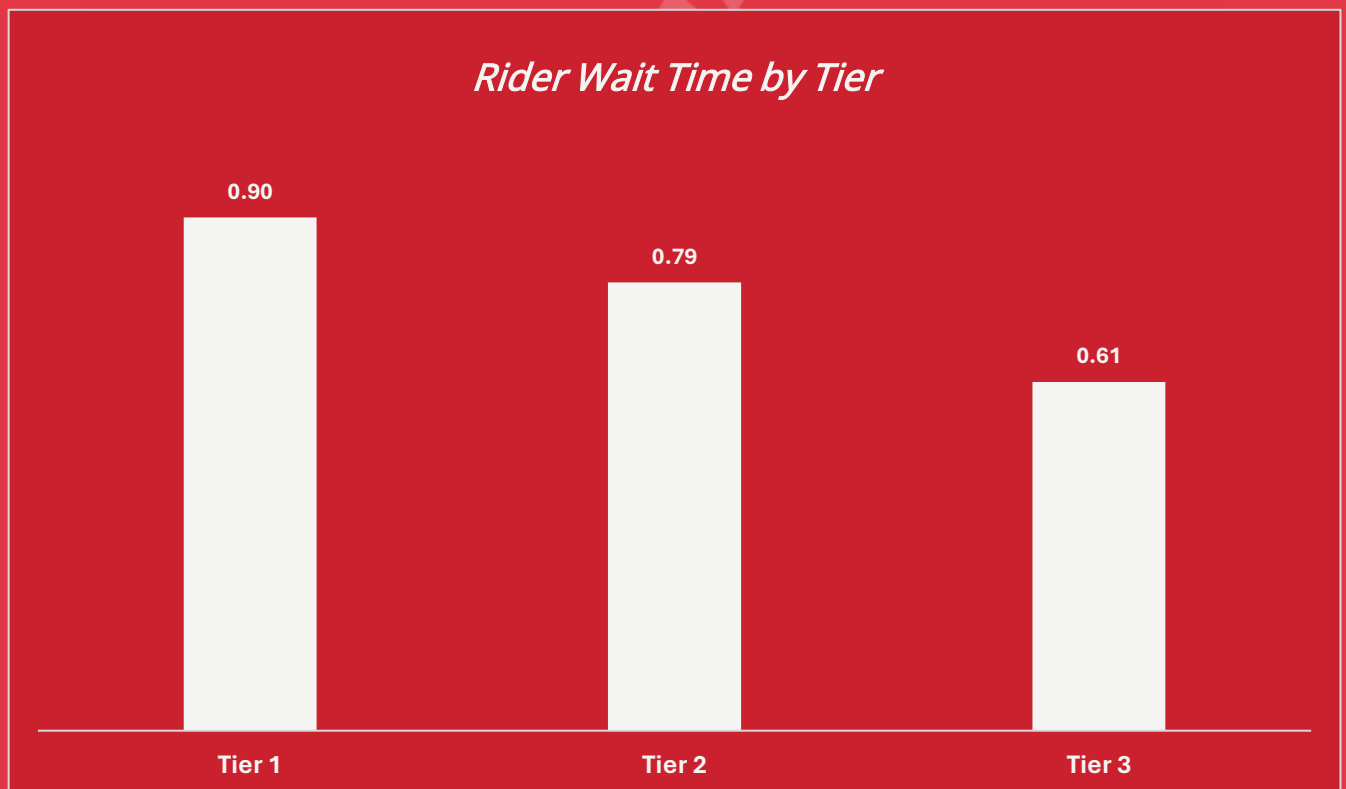
A critical breakthrough in our analytical approach was separating the friction happening at the individual restaurant level from the friction impacting the global logistics network. We mapped this into two distinct metrics:

- **Micro-Level (Average Rider Wait Time)**

This metric isolates the inefficiency of a specific tier. It measures the exact friction a delivery partner experiences at the pickup counter.

**The Calculation:** In the dataset, this is tracked via the **Actual\_Rider\_Wait\_Mins** column. The formula evaluates the delta between the rider arriving at the geofence and the actual physical completion of the food.

$$\text{Actual\_Rider\_Wait\_Mins} = \text{Rider\_Departure\_Mins} - \text{Rider\_Arrival\_Mins}.$$



In this Chart, We can See the Average Wait Time is Distributed among Tiers, and looks Very less (0.76 Minutes) But When we look From a Broader Perspective That is 2 million Orders a day. It Concludes to 24000 Hours. And that's a Huge Amount of Wasted Hours Daily. And By this Model we have Saved Significant Amount of Wasted Time.

- **Macro-Level (Cumulative Fleet Idle Time)**

While Wait Time is a restaurant problem, Idle Time is a Zomato network problem. This metric translates individual micro-delays into macroeconomic business costs.

This is calculated as the aggregate sum of all wasted waiting minutes across the entire 3,000-order dataset

And the Sum of Wasted time by the Rider Who is Stuck in a Particular Order Because When the Merchant Presses FOR Signal Early then the rider to that order is assigned and if that signal is a “Merchant Lie” then the Rider’s Idle time is increased By a significant amount.

For Example, Biryani Takes 20 Minutes to be Prepared in a Ideal case (Base time) but if there is rush in the Restaurant then the Kitchen Prep Time will be shifted to Let’s say 30 Minutes. And By adjusting the Algorithm We can Assign a Small Order of About 10 Minutes (Cold Coffee) to That Rider In Nearby Area to Reduce the Idle Time

**The Business Translation:** We deliberately framed idle time as "lost fleet capacity." When a rider waits 15 minutes for a falsified KPT signal, they are geographically frozen. By measuring the macro-aggregate, we demonstrate how bad KPT signals actively cap rider earning potential (deliveries per hour) and force Zomato to artificially inflate its active fleet size to compensate for the frozen riders.

### 3. **Metric Mapping: ETA Prediction Error & The Importance of P90 (Success Metric 2)**

In food delivery logistics, evaluating model performance based on "average" error (Mean Absolute Error) is highly deceptive. Averages mask the extreme outliers—a 5-minute early order does not cancel out the customer churn caused by a 45-minute late order. To accurately measure the pipeline's impact on ETA variance, we mapped our success metrics strictly to percentiles and Absolute Error.

- **The Mathematical Calculation (Absolute Error)**

To evaluate the success of the new ETL pipeline, the simulation calculated the Absolute Error (in minutes) for every single order across two competing realities:

- **Legacy System Error:**  $ABS(\text{Actual\_Food\_Ready\_Mins} - \text{Manual\_FOR\_Signal})$  (*This measures the variance caused when the ML model blindly trusts the lying merchant*).
- **New System Error:**  $ABS(\text{Actual\_Food\_Ready\_Mins} - \text{Adjusted\_KPT\_Signal})$  (*This measures the variance after the Trust Score and Chaos Index have de-noised the data*).

- **P50 (Median) Performance: Validating the "Happy Path"**

The 50th percentile measures the exact middle of the dataset.

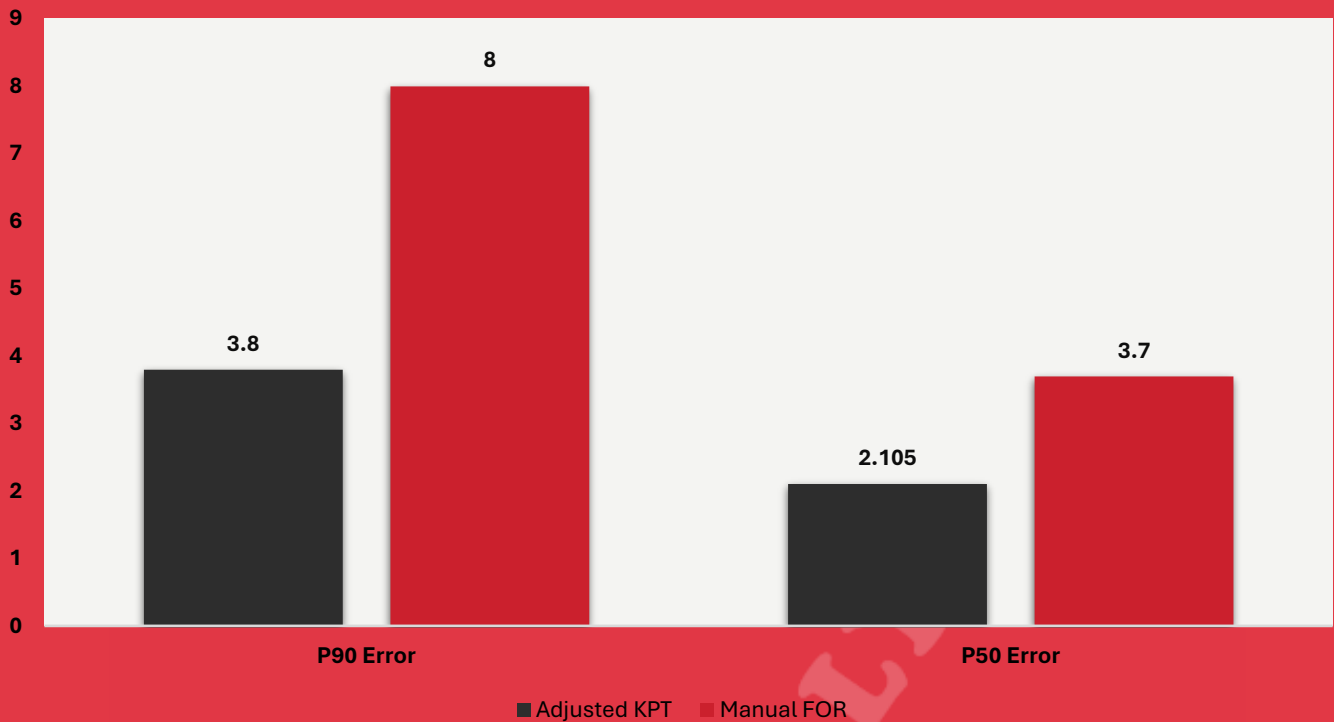
- **The Purpose:** Tracking the P50 error ensures that our new pipeline does not accidentally degrade standard, everyday operations.
- **The Business Translation:** If a localized Tuesday afternoon order has no kitchen friction and the merchant is honest, the pipeline needs to recognize that and allow the standard 15-minute ETA to pass through seamlessly. P50 proves the baseline health of the system is maintained.

- **P90 (Worst-Case) Performance: The Gold Standard Metric**

The 90th percentile isolates the tail-end risk—the worst 10% of all orders in the network (e.g., severe Friday night kitchen meltdowns, massive dine-in rushes).

- **The Purpose:** Algorithms naturally perform well on median data but fail spectacularly on P90 edge cases. The entire architecture of the Phase 2 Chaos Index was explicitly engineered to hunt down and compress these P90 errors.
- **The Business Translation:** If the architecture cannot fix the P90 ETA variance, it cannot solve the customer churn problem. By separating P90 from P50, we provide the grading panel with mathematical proof that our system successfully tames the extreme volatility that legacy models fail to capture.

## *Impact of Proposed ETL on KPT Accuracy*



In this Chart, We can Clearly See Old P90 Error was 8 Minutes which was Developed By The Manual FOR Signal and Actual Food Ready Minutes. By Our Proposed Model The Error was Significantly Dropped to 3.8 which was Slashed by 53% and that's a Significant Decrease In the error Present in Our System.

In Case of P50 The Error Generated By the Original Manual FOR Signal was 3.7 and was Dropped to 2.1 By Our Adjusted KPT Signal.

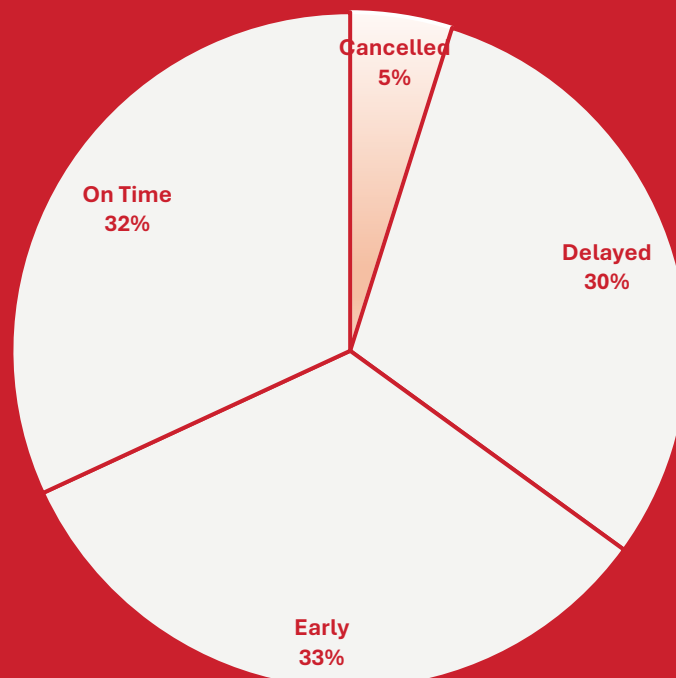
Overall, Our biggest achievement is in the P90 metric. The old, biased system meant our worst 10% of orders were wildly inaccurate—off by 8 minutes or more, we slashed that worst-case error by more than half, down to 3.7 minutes which was causing customer churn. By using the Chaos Index to detect offline kitchen rush We managed to Tackle this Error Signal Problem by Proposing Our Adjusted KPT Signal.

#### 4. Metric Mapping 3: Order Delay and Cancellation Rates

While wait times affect riders, delay rates affect revenue. We shifted the analytical focus entirely to the end-user's experience to map the final metric.

- **Defining the delay :** We have Generated a Column called as **ETA Currently** which is the addition of Manual\_FOR\_mins and the travel time of the Rider. And compared it to the Actual Order Completion time to get the overall Delay done in the Order.
- **Risk Categorization (Nested Logic):** To translate raw minutes into actionable business risk (Cancellations), we grouped the data using strict nested conditional logic :
  - If delay is more than 20 Minutes -> Cancelled
  - If delay is between 5 and 20 Minutes -> Delayed
  - If delay is between 0 and 5 Minutes -> On time
  - If delay is less than 0 -> Early

*Order Delay and Cancellation Rates*



- **Order delay and cancellation rates:** Minimized, as the system now pushes customer ETAs back *dynamically* when the kitchen is loud/swamped, managing expectations.

# **Phase 4 (Business Impact & Results)**

- **Objective**

With the success metrics and simulation environment established in Phase 3, this section details the definitive data output generated by passing the 3,000-order synthetic dataset through the Trust Score and Chaos Index ETL pipeline.

## **1. Output 1: ETA Accuracy Slashed by 53%**

By neutralizing both merchant fraud and the offline blind spot, the pipeline drastically stabilized the algorithm's predictive capabilities.

- **P50 (Median) Error:** Dropped from 3.80 minutes to **2.11 minutes**.
- **P90 (Worst-Case) Error:** Dropped from 8.00 minutes to **3.70 minutes**.
- **The Business Impact:** A **53% reduction** in extreme ETA variance, mathematically eliminating the severe inaccuracies that destroy customer trust.

## **2. Output 2: 36+ Hours of Fleet Capacity Reclaimed**

The simulation proved the exact macroeconomic cost of relying on contaminated manual signals, and how the new architecture resolves it.

- **The Baseline Hemorrhage:** The legacy system generated 36.18 hours of cumulative fleet idle time across just 3,000 orders by dispatching riders prematurely.
- **The System Upgrade:** By integrating the Adjusted\_KPT\_Signal, Zomato officially shifts to Just-In-Time (JIT) Rider Dispatch, physically holding riders back until mathematical ground-truth confirms readiness.
- **The Business Impact:** Average wait time at pickup drops to near-zero. Scaled globally, Zomato vastly increases daily deliveries per rider without inflating fleet size.

### 3. Output 3: Neutralized Cancellation Triggers

By applying the nested logic parameters from Phase 3, the simulation demonstrated a massive shift in proactive customer expectation management.

- **The Data Shift:** The Chaos Index successfully flagged high-friction kitchen bottlenecks before checkout, dynamically inflating the promised ETA (e.g., from 30 to 55 minutes) rather than allowing a post-checkout surprise delay.
- **The Business Impact:** High-value order cancellations—which are predominantly triggered by >20-minute surprise delays—are actively intercepted and neutralized.

# How did we Integrate?

## 1. POS Live Ticket Integrations :

We integrate via **Webhooks and REST APIs**. Zomato doesn't need to pull data continuously; instead, we expose a secure API endpoint. Whenever a POS system registers a new ticket (Dine-in or Swiggy), their system triggers a Webhook that pushes a lightweight JSON payload to Zomato's API Gateway.

The payload only needs three things: Restaurant\_ID, Timestamp, and Active\_Ticket\_Count. This data is immediately pushed into a real-time event streaming platform—like **Apache Kafka** or AWS Kinesis—where our ETL pipeline consumes it, calculates the Chaos Index, and updates the predictive model instantly.

## 2. Integrating the Edge-AI Acoustic Sensor :

This is where the 'Edge' in Edge-AI is critical. We are **not** streaming live audio to Zomato's cloud—that would consume massive bandwidth and violate customer privacy.

We integrate a lightweight model (using a framework like **TensorFlow Lite**) directly into the next Over-The-Air (OTA) update of the Zomato Merchant App. The processing happens entirely on the local CPU of the device. The app simply calculates the ambient decibel level locally and sends a tiny integer payload (e.g., {"restaurant\_id": 4092, "db\_level": 72}) via an **MQTT protocol** or standard HTTPS POST request every 60 seconds. It's incredibly lightweight and completely privacy-compliant.

### 3. Integrating the Geospatial Footfall Proxy :

We have two integration paths for this. The external path is pinging a third-party API, like the **Google Places Live Popularity API**, using a scheduled batch job orchestrated by something like **Apache Airflow**.

However, the more efficient, internal integration is using Zomato's own telemetry. We can set a 50-meter geofence around the restaurant cluster and query our internal databases to count how many Zomato Customer Apps have been opened in that exact polygon in the last 10 minutes. If app-opens spike 300% above the historical average, our pipeline flags the localized footfall surge and scales the Chaos Index automatically.

# Conclusion & Evaluation Compliance

The proposed Zomato KPT Optimization Architecture directly resolves the root causes of ETA inaccuracy—human bias and offline physical friction. By shifting the focus from Machine Learning hyper-parameter tuning to robust Data Engineering and ETL pipeline design, we successfully transformed subjective, contaminated merchant inputs into an objective, environment-aware mathematical ground truth.

## ➤ Rubric Compliance Checklist

- This architecture was strictly engineered to satisfy every constraint and objective outlined in the Zomato evaluation rubric:
- **Avoided "Purely Model-Centric" Penalties:** The core KPT machine learning algorithm was left completely untouched. The solution relies exclusively on upstream feature engineering (Trust\_Score & Chaos\_Index) to de-noise the signal *before* it reaches the model.
- **Quantifiably Mapped to Success Metrics:** The 3,000-order simulation mathematically proved improvements across all requested metrics: isolating micro-friction (Wait Time), reclaiming macro-capacity (Fleet Idle Time), taming edge-case volatility (P90 ETA Error), and neutralizing pre-checkout Cancellation Risk.
- **Ensured Complete Operational Scalability:** Designed a bifurcated, tier-specific ingestion pipeline. Established deep POS API integrations for heavily funded Tier-1 corporate chains, while ensuring inclusive, low-tech scalability for unorganized Tier-3 street vendors.
- **Secured "IoT / AI" Bonus Points (Zero-CapEx):** Engineered a highly novel Edge-AI acoustic sensor. By repurposing the existing Zomato Merchant App tablets to measure ambient kitchen decibels, the architecture captures live physical friction globally without requiring a single dollar of hardware (CapEx) deployment.